A star in disguise?

OBSERVATIONAL ASTRONOMY / STUDENT FIELD NOTEBOOK

Student / group: _____ Date: _____

A compact dot can preserve the history of a larger galaxy.

Ultra-compact dwarf galaxies (UCDs) are compact stellar systems with a fuzzy boundary against massive star clusters. Some may be galaxy centers left behind after outer stars were stripped away. Comparing their properties helps us study how galaxies grow. M60-UCD1 even hosts a massive black hole discovered through detailed follow-up [1,2]; this worksheet cannot establish a black hole or confirm a new UCD.

Why Gaia can record non-stars

Gaia scanned the sky and selected compact concentrations of light for small measurement windows. This favors star-like light patterns, not necessarily individual stars: compact galaxies, bright galaxy centers, and quasars (active galaxy centers) can be detected too. Diffuse light is less favored. Departures from a stellar image or motion fit are clues, but can also arise from blends or measurement problems [3].

Do not confuse	Numerical context and meaning
Pixel sampling	0.0589 arcsec along scan; 0.1768 arcsec across. Pixel size is not resolving power [4].
Onboard pair resolution	About 0.23 arcsec along and 0.70 arcsec across for equal-brightness pairs in the detection study. Not a UCD-radius cutoff [3].
Catalog completeness	EDR3 pair losses start below about 1.5 arcsec, becoming severe below 0.7 arcsec. DR3 uses the EDR3 astrometric solution [5].
Positional precision	A source center can be measured more precisely than its blurred image width. This does not resolve its individual stars.

Your job

Compare known objects, review two published candidates, and record what you see and cannot tell. A compact image, no detected motion, or high flux excess alone never confirms a UCD. Missing astrometry is not zero. The Gaia acronym UCD can also mean an ultracool stellar dwarf - a different topic.

Use the companion HTML offline, or complete this PDF on paper. Optional online practice is on page 6. No code or website is required. References are on page 6.

01 / Identify and compare



M60-UCD1



M59-UCD3



Foreground comparison

DR11 close cutouts: 33.536 arcsec square each, 0.262 arcsec/pixel. Centers are requested Gaia coordinates; images have no drawn marker. The HTML also supplies wider context; paper-only reviewers should note that missing context.

Gaia DR3	M60-UCD1	M59-UCD3	Foreground comparison
Source ID (text)	3927659910436411136	3928413763095214592	3928418917055971968
G magnitude	18.115	17.850	17.129
BP-RP	1.394	1.363	1.114
Raw flux excess C	4.654	6.135	1.220
Parallax (mas)	-0.079 +/- 0.395	0.370 +/- 0.417	0.788 +/- 0.076
pmra (mas/year)	0.420 +/- 0.475	0.096 +/- 0.491	10.984 +/- 0.090
pmdec (mas/year)	-0.547 +/- 0.432	-0.361 +/- 0.488	-50.070 +/- 0.082

Parallax is an annual apparent shift; proper motion is a drift across the sky. One milliarcsecond (mas) = 0.001 arcsec. pmra includes the cosine of declination; pmdec is north-south motion. Values are measurement +/- uncertainty; negative parallax can arise from noise, and missing is not zero.

Q1. Explain why searching for UCDs can tell us something about how galaxies grow. Give one possible origin, and one thing this exercise cannot establish.

Q2. Inspect M60-UCD1 and the foreground comparison at the available image scales (both close and context in HTML; close only in the PDF). Record their exact Gaia identifiers and one visible similarity or difference. Is the image alone enough to separate a star from a UCD? Explain.

Q3. Calculate abs(parallax) / parallax_error for both. State which has convincing foreground evidence and cite the motion measurements as well. Explain why no detected motion in M60-UCD1 does not confirm a UCD.

02 / Size and uncertainty

Small-angle relation: angular radius (arcsec) = 206265 x radius (pc) / distance (pc). Sizes/distances use parsecs (pc); one megaparsec (Mpc) is 1,000,000 pc. The result is a half-light radius, not a diameter or measured image width. On paper, calculate the 5 and 50 Mpc examples instead of moving the slider.

Q4. M60-UCD1 has a published half-light radius of 24.2 pc at an adopted host distance of 16.5 Mpc. Calculate its angular half-light radius using 206265 x radius_pc / distance_pc. Move the activity's distance slider to compare 5 and 50 Mpc. What changes, and what stays physically the same?

Q5. Why is Gaia's 0.0589-arcsecond along-scan pixel size not a universal resolving limit? Explain angular resolution versus positional precision. Must a compact galaxy be perfectly point-like to be detected?

Raw excess C = (I_BP + I_RP) / I_G, using blue, red, and broad-band fluxes, not magnitudes. Renormalized unit weight error (RUWE) measures astrometric fit quality; excess noise is additional angular scatter, not size. High values can have several causes. These six examples are deliberately selected.

Q6. Compare G and raw flux excess for the foreground star, M60-UCD1, and M59-UCD3. Give two alternative explanations for unusual flux excess or poor stellar fit. Why can six selected examples not establish a threshold or contamination rate?

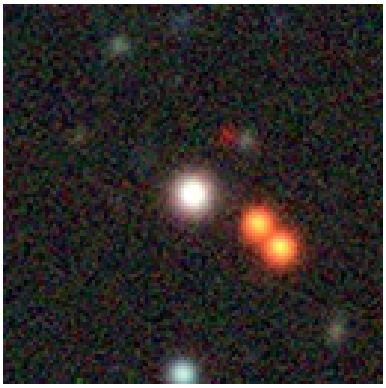
M59cO - compact-system boundary case	Saved Gaia DR3 record
Source ID	3928425445406259456
RA, Dec (degrees)	190.48055767, 11.66771948
Parallax / proper motion / RUWE	Not supplied / Not supplied / Not supplied

Q7. Inspect M59cO's catalog panel. Does its missing parallax mean zero parallax? What can and cannot be concluded about its motion?

03 / Published candidate A

J124352.42+112534.2

Gaia ID 3927645994742413056 / RA 190.96839498, Dec 11.42614378 deg



Close: 33.536 arcsec square



Context: 204.800 arcsec square

Legacy Survey DR11; close 0.262 arcsec/pixel, context 0.8 arcsec/pixel. Both centered on Gaia position, not proof of a correct match. Inspect the unmarked image center; the HTML also offers a marker.

Saved Gaia DR3	Value
G / BP-RP / raw excess C	19.376 / 1.224 / 3.495
Parallax (mas)	0.314 +/- 0.641
pmra / pmdec (mas/year)	0.267 +/- 0.881 / -0.363 +/- 0.744
RUWE / excess noise (mas)	1.791 / 3.851

Flags: Y=yes / N=no / U=uncertain / NA=not assessable. Circle one for each.

Visibly resolved	Y N U NA
Off-center / possible mismatch	Y N U NA
Close to host light	Y N U NA
Blend with a nearby source	Y N U NA
Too faint to assess	Y N U NA
Image problem / artifact	Y N U NA

R1. Image evidence and uncertainty

R2. Astrometry and alternative explanations

R3. Recommendation reason and next evidence

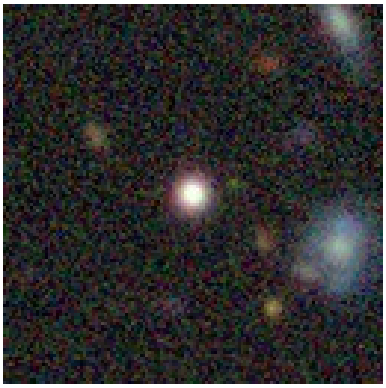
Recommendation: retain for instructor review / possible contaminant / insufficient information

Published candidate, not confirmed. Liu 2015 table6.dat [6]; inherited comparison Fahrion 2019 [7]. Original labels differ; approved reference state is candidate. Assumed M60 distance is conditional. External quasar, galaxy, and non-single-star memberships are not checked. Color alone cannot identify a young cluster or H II region (gas ionized by young stars).

04 / Published candidate B

J124524.02+110535.2

Gaia ID 3927574728348720512 / RA 191.35006737, Dec 11.09309767 deg



Close: 33.536 arcsec square



Context: 204.800 arcsec square

Legacy Survey DR11; close 0.262 arcsec/pixel, context 0.8 arcsec/pixel. Both centered on Gaia position, not proof of a correct match. Inspect the unmarked image center; the HTML also offers a marker.

Saved Gaia DR3	Value
G / BP-RP / raw excess C	19.660 / 1.280 / 2.574
Parallax (mas)	-1.139 +/- 0.708
pmra / pmdec (mas/year)	-0.294 +/- 0.880 / 1.579 +/- 0.778
RUWE / excess noise (mas)	1.563 / 4.236

Flags: Y=yes / N=no / U=uncertain / NA=not assessable. Circle one for each.

Visibly resolved	Y N U NA
Off-center / possible mismatch	Y N U NA
Close to host light	Y N U NA
Blend with a nearby source	Y N U NA
Too faint to assess	Y N U NA
Image problem / artifact	Y N U NA

R1. Image evidence and uncertainty

R2. Astrometry and alternative explanations

R3. Recommendation reason and next evidence

Recommendation: retain for instructor review / possible contaminant / insufficient information

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05 / Reflect, explore, submit

Q8. Identify one uncertainty or possible mismatch and the single most useful next observation or catalog check. Separate observation from interpretation.

Optional online steps - no code

Gaia Archive: open gea.esac.esa.int/archive/; select DR3 and object/position search. Use the exact source ID if supported, otherwise the supplied decimal-degree position and a 5 arcsec matching radius. Check the exact returned ID, values, errors, and units. Do not turn missing fields into zero.

Gaia Visualization: use its sky view for context and supported plots. Exact object identification may require the Archive or Aladin. The built-in HTML comparison plot is the required plot; a parallax-based stellar 3D view cannot give a UCD distance.

Aladin Desktop: install the version for your operating system from CDS. Enter the coordinates; load an optical image and Gaia DR3 catalog I/355/gaiadr3 from VizieR. Select the matching table row and toggle catalog symbols. Record the image survey/release and consider image epoch for moving stars.

Legacy Survey: use each object's viewer link in the HTML; confirm the DR11 image layer, not model/residual. Inspect close and wider views, coordinates, scale, host light, and defects. A blank tile is unavailable imaging; RGB display colors are not calibrated colors.

Live GUI labels vary and were not fully tested. The full step-by-step guide is embedded in the HTML. Core work is offline. No Windows rehearsal or student installation is required.

Submission checklist

Record your identifier; answer Q1-Q8 and both candidate reviews; distinguish observations from interpretation. Export JSON to resume and text to read/submit, or submit this PDF. Export before closing: local browser storage is best-effort. Nothing uploads automatically, and student reviews never become confirmed/training labels without instructor adjudication.

References and provenance

[1] Strader et al. 2013 / M60-UCD1: https://arxiv.org/abs/1307.7707

[2] Seth et al. 2014 / M60-UCD1 black hole: https://arxiv.org/abs/1409.4769

[3] de Bruijne et al. 2015 / onboard detection: https://arxiv.org/abs/1502.00454

[4] Gaia Collaboration 2016 / mission instrument: https://arxiv.org/abs/1609.04153

[5] Fabricius et al. 2021 / EDR3 validation: https://arxiv.org/abs/2012.06242

[6] Liu et al. 2015 / published candidates: https://doi.org/10.1088/0004-637X/812/1/34

[7] Fahrion et al. 2019 / comparative compilation: https://doi.org/10.1051/0004-6361/201834941

[8] Sandoval et al. 2015 / M59-UCD3: https://arxiv.org/abs/1506.08828